



THE Z-DISTRIBUTION

INFERENTIAL STATISTICS

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TOPIC OUTLINE

Point Estimate

Confidence Interval

z-Distribution



POINT ESTIMATE



POINT ESTIMATE

A point estimate is a single value (statistic) derived from sample data that serves as the "best guess" for an unknown population parameter.

$\bar{x}$ is a point estimate of μ

s^2 is a point estimate of σ^2

example

A factory produces resistors labeled as **100 Ω** , but due to manufacturing variations, the actual resistance varies. An engineer takes a random samples of 30 resistors and calculated the average resistance of the sample, $\bar{x} = \mathbf{101.2 \Omega}$.

The point estimate for the true mean resistance μ of all resistors produced is **101.2 Ω** .

No tolerance $\rightarrow$ $\left. \begin{array}{l} 101.3 \Omega \\ 101.2 \Omega \end{array} \right\} \text{NG} - \text{No Good}$
Good must be exactly 101.2 Ω



CONFIDENCE INTERVAL

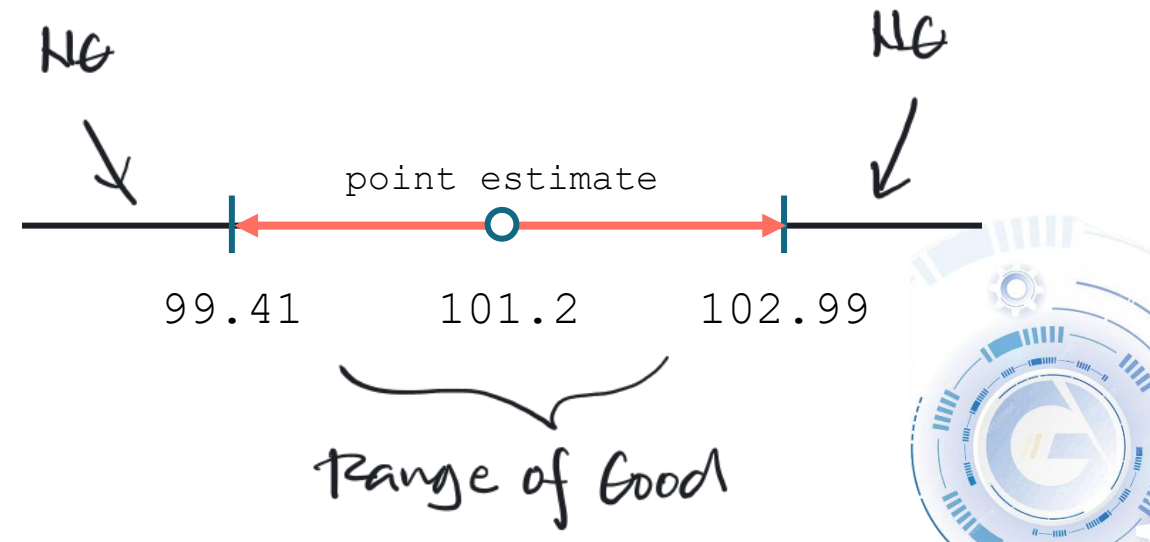


CONFIDENCE INTERVAL

A confidence interval is a range of values, derived from sample data, that is likely to contain the true value of an unknown population parameter (e.g., μ, σ^2).

example

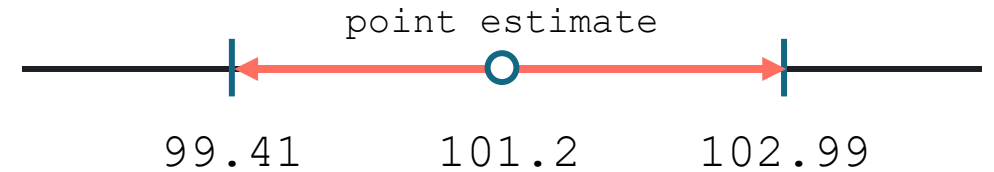
A factory produces resistors labeled as **100 Ω** , but due to manufacturing variations, the actual resistance varies. An engineer takes a random samples of 30 resistors and calculated a 95% confidence interval for the true mean resistance, **95% CI = [99.41 Ω , 102.99 Ω]**



CONFIDENCE INTERVAL

A confidence interval is a range of values, derived from sample data, that is likely to contain the true value of an unknown population parameter (e.g., μ, σ^2).

example



We are 95% confident that the true mean μ of all resistors produced lies between 99.41 Ω and 102.99 Ω .

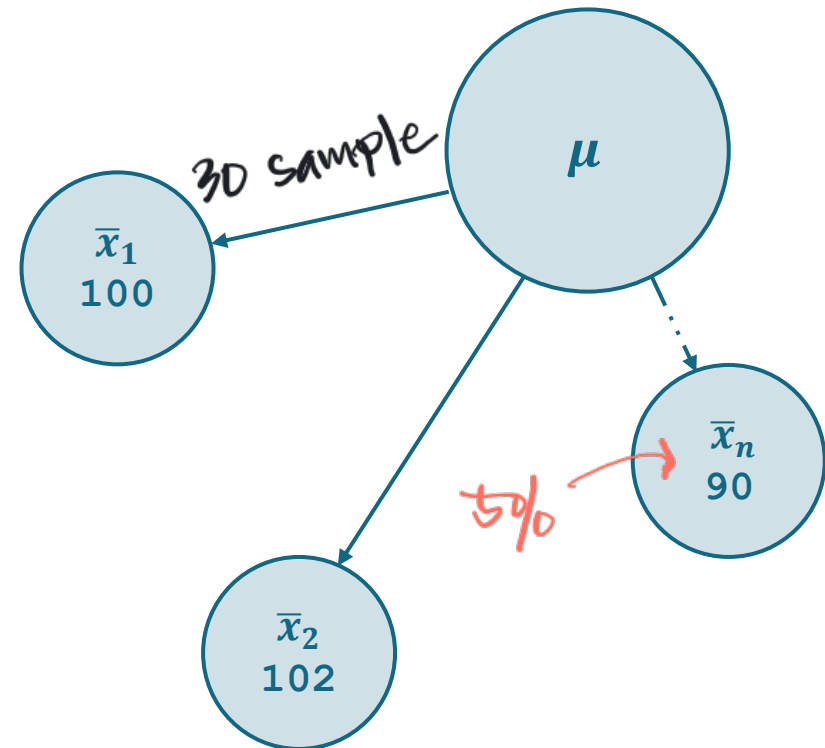
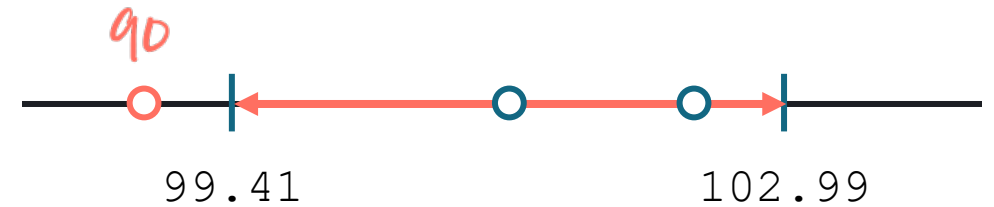


CONFIDENCE LEVEL

Confidence levels (e.g., 90%, 95%, 99%) describe the method's reliability over many samples.

A 95% confidence level means that if the same sampling process were **repeated** many times, approximately 95% of the calculated CIs would contain the true population parameter (e.g., μ , σ^2).

example



CONFIDENCE LEVEL

Confidence levels (e.g., 90%, 95%, 99%) describe the method's reliability over many samples.

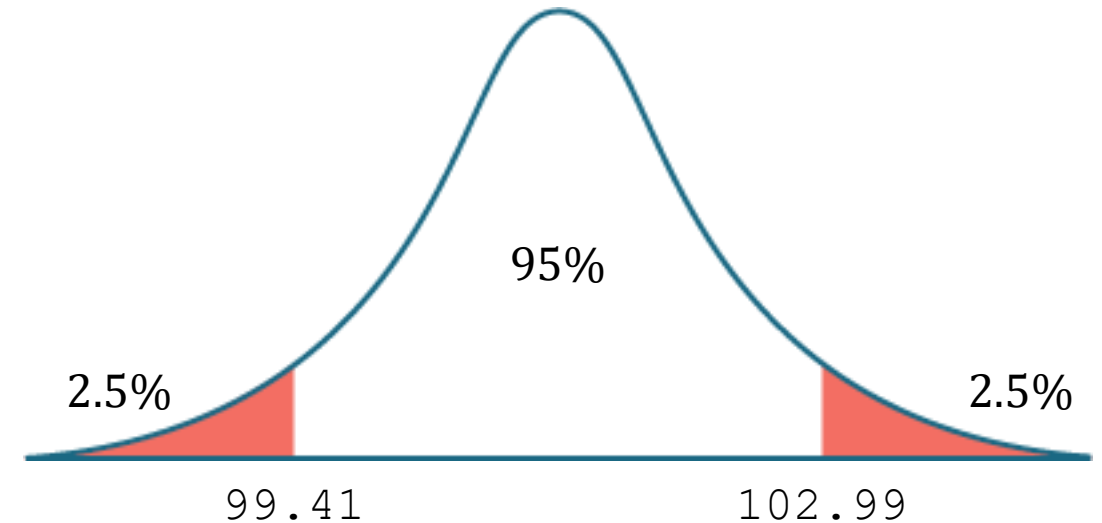
Formula

$$\text{confidence level} = 1 - \alpha$$

where

$$0 \leq \alpha \leq 1$$

95% Confidence Level



$$\alpha = 0.05$$

95% confidence calculus exam
between 5 and 7



CONFIDENCE LEVEL

Confidence levels (e.g., 90%, 95%, 99%) describe the method's reliability over many samples.

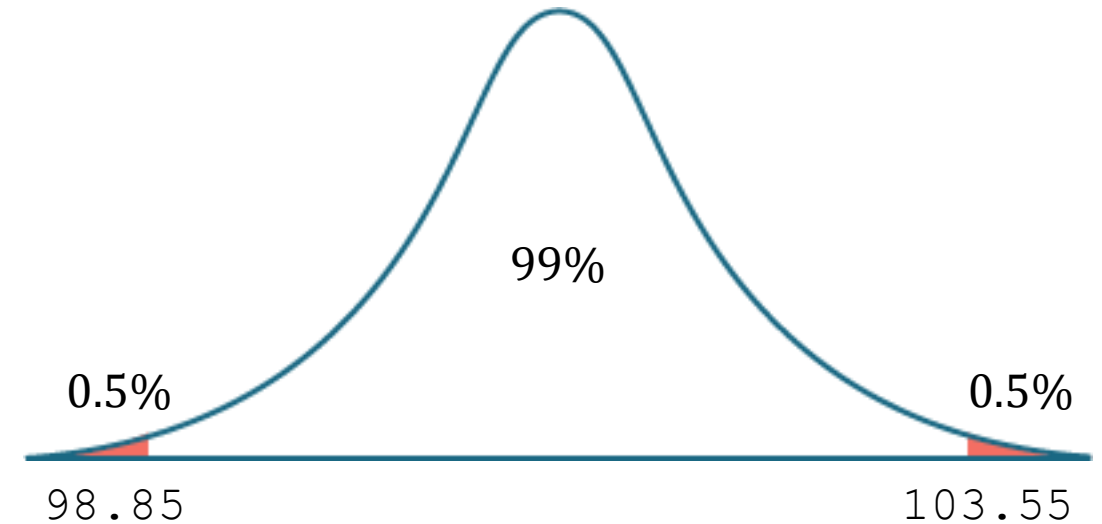
Formula

$$\text{confidence level} = 1 - \alpha$$

where

$$0 \leq \alpha \leq 1$$

99% Confidence Level



$$\alpha = 0.01$$

99% confidence calculus exam
between 3 and 8



CONFIDENCE LEVEL

Confidence levels (e.g., 90%, 95%, 99%) describe the method's reliability over many samples.

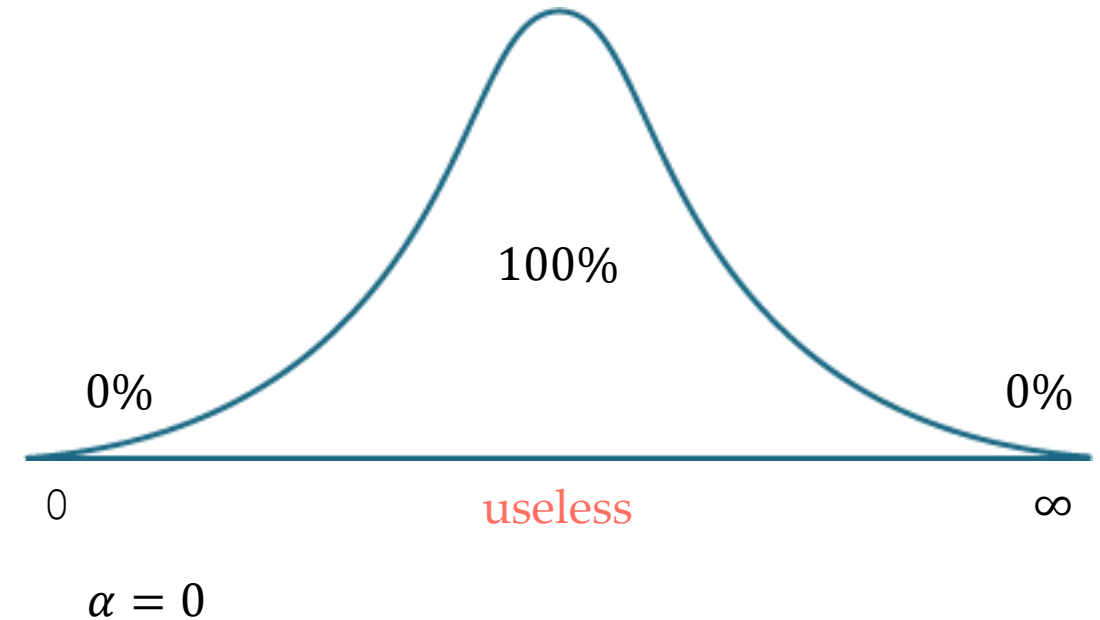
Formula

$$\text{confidence level} = 1 - \alpha$$

where

$$0 \leq \alpha \leq 1$$

100% Confidence Level



100% confidence calculus exam
between 0 and 10



Z-DISTRIBUTION



Z-DISTRIBUTION

The z-distribution is used to calculate the confidence interval when:

1. The population variance (σ^2) is known.
2. The sample size is large ($n \geq 30$).

Formula

$$CI = \bar{x} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

where

$\bar{x}$ = sample mean

$z_{\alpha/2}$ = z-statistic

$\frac{\sigma}{\sqrt{n}}$ = standard error

z-table

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990

$$z_{0.025} = 1.9 + 0.06$$

$$z_{0.025} = 1.96$$

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Formula

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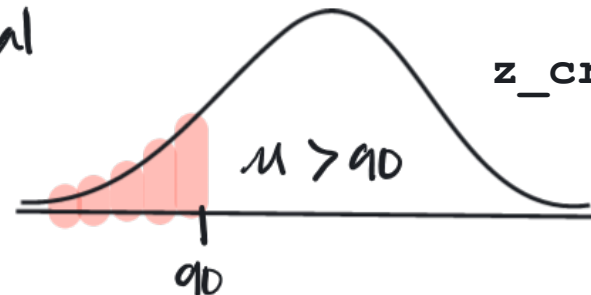
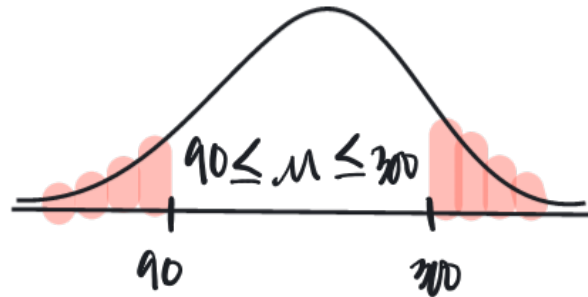
where

$\bar{x}$ = sample mean

$z_{\alpha/2}$ = z-statistic

$\frac{\sigma}{\sqrt{n}}$ = standard error

z-critical



syntax

Percent point function `norm.ppf()` returns the critical **z** value for a given probability ($1-\alpha$).

```
from scipy import stats
```

Two-Tailed Test

```
z_critical = stats.norm.ppf(1-alpha/2)
```

One-Tailed Test

```
z_critical = stats.norm.ppf(1-alpha)
```

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Formula

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$\frac{\sigma}{\sqrt{n}}$ = standard error

syntax

Interval function `norm.interval()` returns a tuple (lower_limit, upper_limit) representing the confidence interval for z-interval.

```
from scipy import stats
```

```
ci_lower, ci_upper =
```

```
stats.norm.interval(
```

```
    confidence = confidence_level,
```

```
    loc = sample_mean,
```

```
    scale = standard_error
```

```
)
```



EXERCISE

A power company measures the voltage output (in volts) of a batch of transformers. The population standard deviation (σ) is known to be **0.5** volts. A random sample of **30** transformers are tested, and their voltage outputs are recorded in "transformer-voltage" data . Calculate a **95% confidence interval** for the true mean voltage output (μ) of all transformers.

Solution

Sample mean

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\bar{X} = \frac{119.8 + 120.3 + \dots + 120.4}{30}$$

$$\bar{X} = 120$$



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Solution

z-statistic

$\alpha = 0.05$ for 95% confidence level

$$z_{\alpha/2} = z_{0.025} \rightarrow 1 - 0.025 = 0.975$$

z-table

from z-table

$$z_{0.025} = 1.96$$



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$$\bar{x} = 120$$

$$z_{0.025} = 1.96$$

Solution

Confidence interval

$$CI = \bar{x} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

0.025

$$CI = 120 \pm 1.96 \left(\frac{0.5}{\sqrt{30}} \right)$$

$$CI = [119.82, 120.18]$$

ans

95% Confidence that the true mean voltage output lies between 119.82V and 120.18V



LABORATORY

